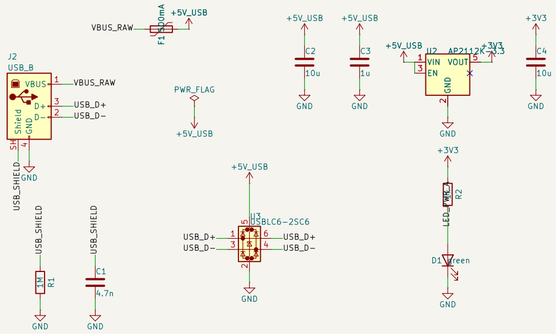
USB INPUT, PROTECTION, 3.3 V



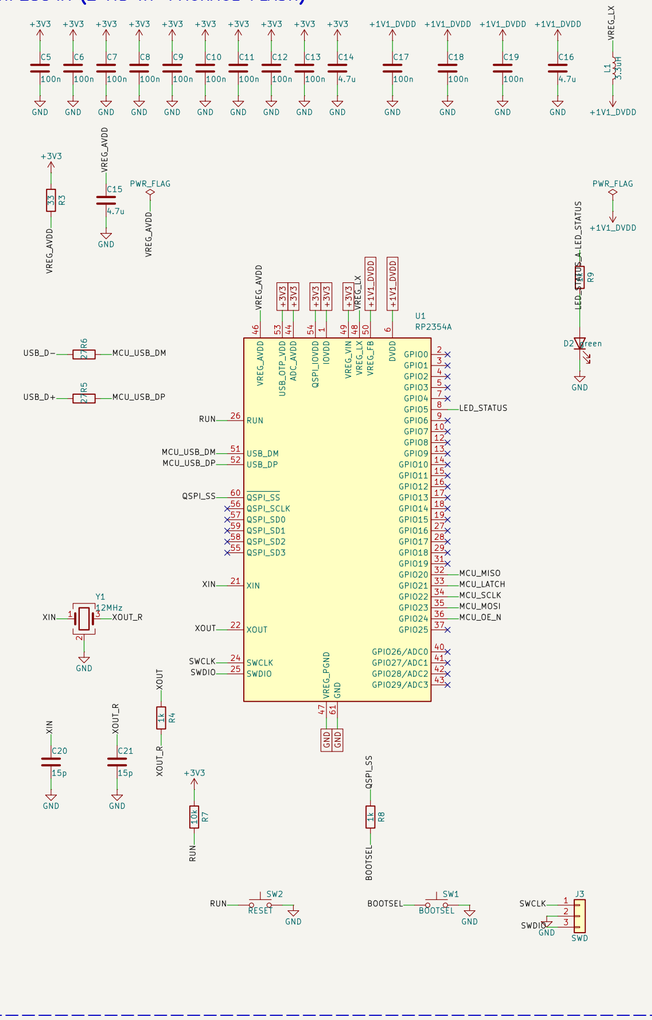
NOTES

Isolated serial link (RP2354A SPI0 / PIO): SPI mode 0, MSB first, <= 1 MHz, 40 clocks per transfer.
Pulse LATCH low >= 1 us, then clock 5 bytes. LATCH low: the 74HCT165s sample the meter outputs.
LATCH rising edge: the 74HC595 outputs take the last two bytes clocked in.
/OE (GPIO24) high = command outputs off (the default while the MCU is unpowered or in reset).

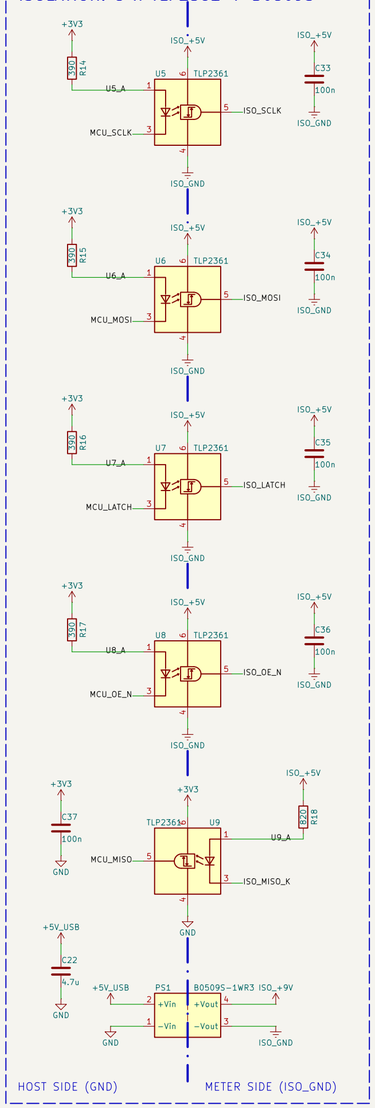
MISO byte0 U10: 2x10^2 PRINT_PULSE RANGE_1 1x10^2 2x10^3 RANGE_4 4x10^2 8x10^2
byte1 U11: FUNC_B POL- RANGE_2 FUNC_A 1x10^4 4x10^3 POL+ 8x10^3
byte2 U12: 4x10^4 2x10^4 4x10^0 1x10^0 DATA_CAN_CHANGE 8x10^4 8x10^0 1x10^1
byte3 U13: 4x10^5 2x10^1 2x10^5 4x10^1 PRINT_LEVEL 8x10^5 8x10^1 1x10^6
byte4 U14: 1x10^5 1x10^3 OVERLOAD 2x10^0 1 0 1 0
MOSI byte3 U16: – – LOCKOUT_N FUNC_43 SAMPLE_PULSE FUNC_42 RATIO_N CONTACT_SAMPLE
byte4 U15: REMOTE_LED RANGE_4 RANGE_2 RANGE_1 INTEG_2 AUTORANGE_INH INTEG_1 INTEG_4
(bit7 first; 1 / 0 = U14 link-check inputs tied high / low)

Meter levels (7075 manual sect. 9): outputs '1' = 2.4..6 V from 6k, '0' sinks 10 mA;
inputs '1' = 2.4..5 V, '0' < 0.5 V at 5 mA. Series 10k on every meter output.
ISO_GND = SKB pin 37 (meter earth / logic 0). Net labels of the same name are connected.

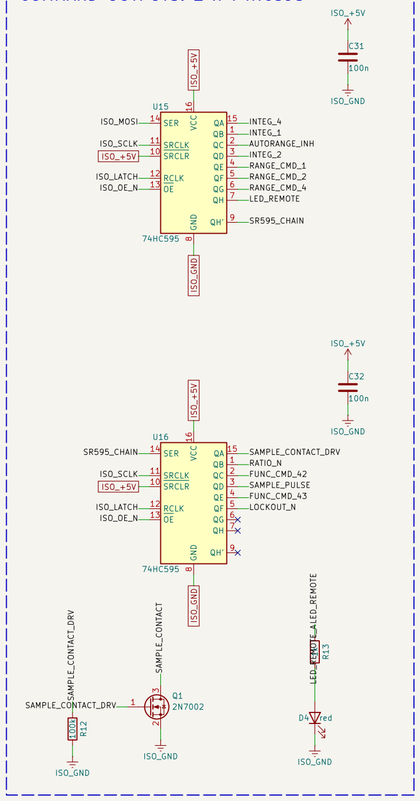
RP2354A (2 MB IN-PACKAGE FLASH)



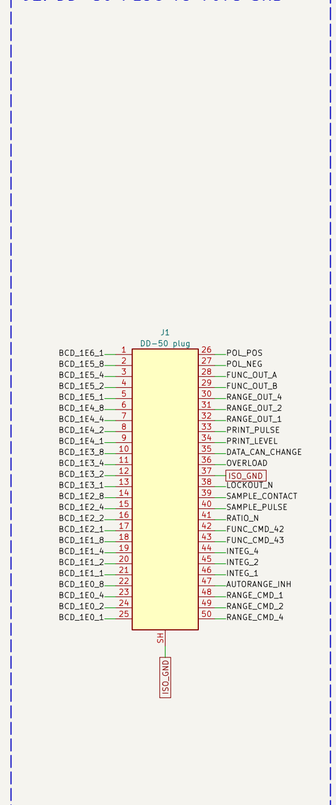
ISOLATION: 5 x TLP2361 + B0509S



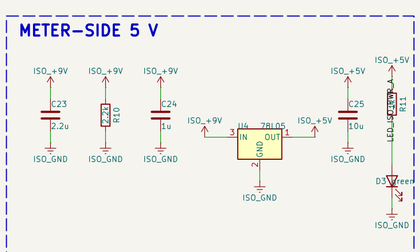
COMMAND OUTPUTS: 2 x 74HC595



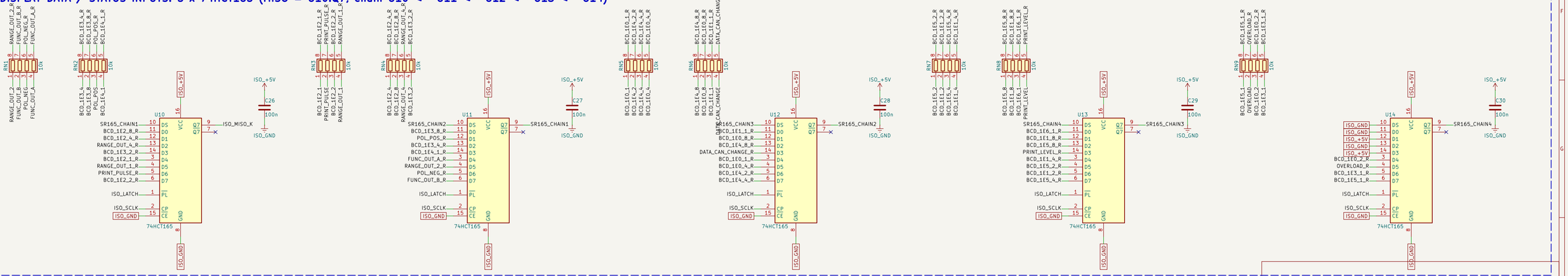
J1: DD-50 PLUG TO 7075 SKB



METER-SIDE 5 V



DISPLAY DATA / STATUS INPUTS: 5 x 74HCT165 (MISO = U10.Q7; chain U10 <- U11 <- U12 <- U13 <- U14)



Net labels with the same name are connected
Generated from the SKiDL netlist by scripts/gen_schematic.py

Sheet: /
File: solartron_7075_interface.kicad_sch

Title: Solartron 7075 USB interface (SKiDL)

Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.6 | Id: 1/1